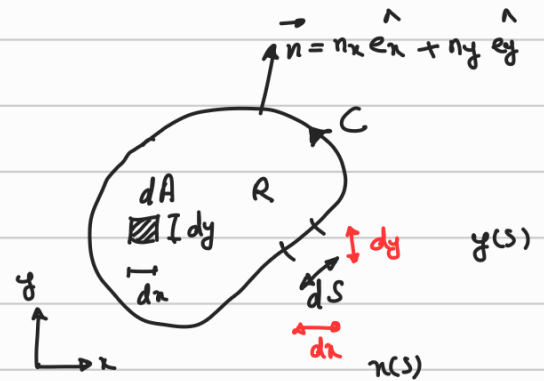


Green's Theorem:

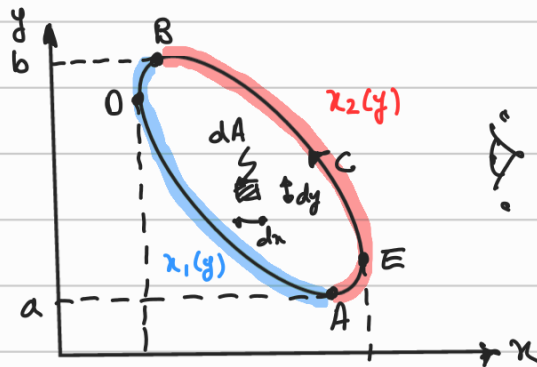
$$P(x, y), \quad \frac{\partial P}{\partial x}, \quad \frac{\partial P}{\partial y}$$

$$\int_R \frac{\partial P}{\partial x} dA = \int_C P dy = \int_C P n_x ds \quad (*)$$

$$\int_R \frac{\partial P}{\partial y} dA = - \int_C P dx = \int_C P n_y ds \quad (**)$$



Proof (*):



$$\int_R \frac{\partial P}{\partial x} dA = \int_a^b \int_{x_1(y)}^{x_2(y)} \frac{\partial P}{\partial x} dx dy$$

$$\int_a^b \int_{x_1(y)}^{x_2(y)} \frac{\partial P}{\partial x} dx dy = \int_a^b P(x, y) \Big|_{x_1(y)}^{x_2(y)} dy$$

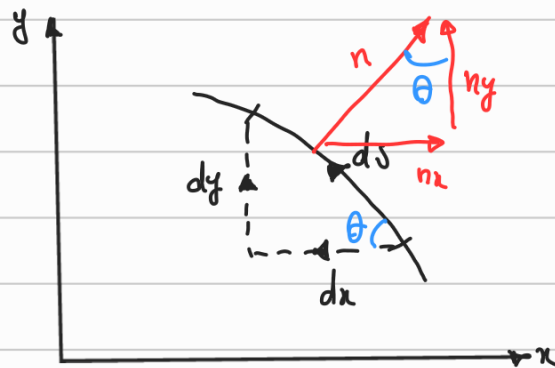
$$= \int_a^b \{ P(x_2(y), y) - P(x_1(y), y) \} dy$$

$$= \int_a^b P(x_2(y), y) dy - \int_a^b P(x_1(y), y) dy$$

$$= \int_{AEB} P dy - \int_{ADB} P dy = \int_{AEB} P dy - \left(- \int_{BDA} P dy \right)$$

$$= \int_{AEB} P dy + \int_{BDA} P dy = \int_C P dy$$

$$\sin \theta = \frac{n_x}{n} = n_x \quad \cos \theta = \frac{n_y}{n} = n_y \quad (1)$$



$$\sin \theta = \frac{dy}{ds} \quad (2)$$

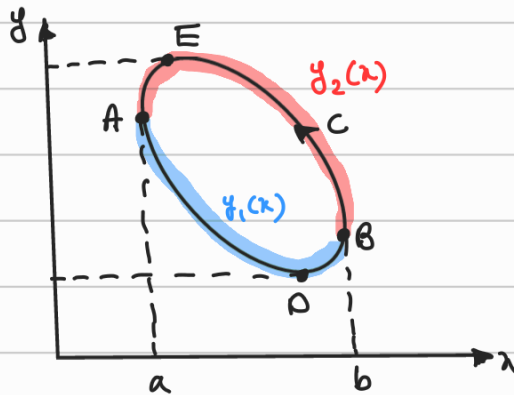
$$\cos \theta = -\frac{dx}{ds}$$

Comparing (1) and (2): $n_x = \frac{dy}{ds}$, $n_y = -\frac{dx}{ds}$

$$\int_C P dy = \int_C P \frac{dy}{ds} ds = \int_C P n_x ds$$

Proof (***):

• A •



$$\int_R \frac{\partial P}{\partial y} dA = \int_a^b \int_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y} dy dx$$

$$\int_a^b \int_{y_1(x)}^{y_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b P(x, y) \Big|_{y_1(x)}^{y_2(x)} dx$$

$$= \int_a^b \{ P(x, y_2(x)) - P(x, y_1(x)) \} dx = \int_a^b P(x, y_2(x)) dx - \int_a^b P(x, y_1(x)) dx$$

$$= \int_{AEB} P dx - \int_{AOB} P dx = - \int_{BEA} P dx - \int_{AOB} P dx = - \int_C P dx$$

$$-\int_C p \, dx^{x(s)} = -\int_C p \frac{dx}{ds} \, ds = -\int_C p (-ny) \, ds = \int_C p ny \, ds$$